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$$f(x) = \ln(x^3 + \sqrt{x^6 + 1})$$

$$f'(x) = \frac{1}{x^3 + \sqrt{x^6 + 1}} \cdot \left(3x^2 + \frac{3x^5}{\sqrt{x^6 + 1}} \right)$$

$$f'(x) = \frac{1}{x^3 + \sqrt{x^6 + 1}} \cdot \left(\frac{3x^2(\sqrt{x^6 + 1}) + 3x^5}{\sqrt{x^6 + 1}} \right)$$

$$f'(x) = \frac{1}{x^3 + \sqrt{x^6 + 1}} \cdot \left(\frac{3x^2(\sqrt{x^6 + 1} + x^3)}{\sqrt{x^6 + 1}} \right)$$

$$f'(x) = \frac{3x^2}{\sqrt{x^6 + 1}}$$

References